

Design and Verilog Implementation of a Byte-Enabled RAM Architecture for Efficient Data Access

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Abstract—Designs for collective digital systems require efficient access to memory. Examples of digital systems include: processor, embedded systems, and systems-on-chip (SoCs). Standard methods of designing RAM modules typically involve writing and reading a single complete data word. However, this often leads to the overwriting of currently existing data already stored in the RAM module. Therefore, these systems can consume significant amounts of energy and inefficiently utilize the digital system's available bandwidth. All digital systems are therefore designed with this form of data loss issue in mind. So, for instance, while designing a CPU one must pay particular heed to the RAM module for new and modified data being written; to ensure no data in the RAM module will be lost because of the newly written data. In the body of this work I will describe a technique for designing RAM that can be written and read in byte mode opposed to word mode. This will be accomplished using only a single instance of a 32-bit data word, and designing this uses Xilinx Vivado.

Keywords—*Synchronous Random Access Memory (SRAM), Development of Very-Large-Scale-Integration (VLSI), Data Storage Systems Warehouse, Very High-Speed Integrated Circuits (VHDL), Digital Systems Design, Embedded Systems, Systems on Chip (SoC), Digital Memory Design, Automated Memory Access, Hardware Description Languages (HDL), Partial Updating of Words, Effective Storage of Data, Memory Control Logic, Field-Programmable Gate Arrays (FPGA), Simulation and Verification and VLSI Silicon Systems. High-Quality Computing Chips.*

I. INTRODUCTION

Memory is one of the most important parts of modern digital technology. This includes everything from small embedded controllers to really large but powerful processors. There is a lot of demand to create more complex and better performing systems, which means that memory operations will affect

system speed, system power use, and system reliability. RAM can do work with fixed word lengths (for example 16 bit long or 32 bit long), so if you want to modify only a portion of a data item in RAM (for example only part of a word), there will be multiple accesses of RAM before that data is completely written back into memory. When developing a simple design that can handle mixed length data (for example variable length data types) there is not an efficient way to update memory when only a portion of a word is to be modified. Most of today's devices based on microcontrollers are going to access data as a byte through their communication interface or through memory mapped I/O devices. With traditional RAM technology, if a microcontroller wants to write 1 byte of information into RAM, the whole word must be completely written back before any access will be done again to print back many accesses before accessing the data; thus, using many (lots) of access times in this regard increases the number of switch events and thus power consumption.

The new architecture plan for "Byte Enabled RAM" will solve the problem of allowing only the bytes to be used for writing memory with specific byte enabled signals being used for writing only the newly needed memory.

This paper details both the design and implementation of a byte enabled RAM architecture using Verilog HDL (hardware description language). It will detail how to create an efficient byte write operation and a byte write to a word size memory system. Simulation and synthesis of this design has been performed to validate the use of this type of architecture in modern day digital system designs requiring partial memory update.

II. LITERATURE SURVEY

Memory architecture design for Field Programmable Gate Arrays (FPGAs) has undergone significant evolution in recent years, driven by the increasing demand for fine-grained, high-efficiency data access in embedded and real-time computing systems. Contemporary applications such as image processing, machine learning inference, and embedded control systems require memory subsystems capable of selectively manipulating individual bytes rather

than entire words, thereby reducing redundant memory transactions and improving overall computational throughput. Against this backdrop, researchers have explored a wide range of memory design strategies aimed at achieving optimal trade-offs among speed, area, and power consumption within FPGA-based platforms.

A foundational contribution to this domain is presented in [1], where a byte-enabled memory architecture implemented in Verilog for FPGA platforms is proposed, enabling selective byte-level data access that eliminates unnecessary full-word write operations. This granularity not only reduces effective memory bandwidth usage but also contributes to lower dynamic power dissipation. Complementing this, [2] introduces a synchronous dual-port RAM design that supports simultaneous read and write transactions within a single clock cycle, significantly enhancing throughput for bandwidth-intensive applications such as graphics rendering. However, this architecture lacks byte-level access control, a limitation that byte-enable-based designs directly resolve. Similarly, [3] presents a Verilog HDL-based, area-efficient DDR SDRAM controller targeting large-scale FPGA deployments, prioritizing maximum data bandwidth while maintaining a compact hardware footprint suitable for resource-constrained environments.

Power and energy efficiency represent equally critical dimensions in modern memory design. In [5], an energy-efficient SRAM-based memory decoder tailored for AI accelerator platforms is described, demonstrating that architectural optimization of the memory decoder stage alone can yield substantial reductions in dynamic power without sacrificing computational throughput. Building upon this concept, [6] presents the co-design of a 4-bit ALU integrated with a compact RAM module, illustrating that ultra-low-power computing systems can be realized through tight coupling of processing and memory resources. These works collectively underscore the importance of power-aware memory design, particularly in battery-operated and thermally constrained FPGA applications where energy budgets are strictly bounded.

Advanced memory exploration and non-volatile memory technologies further expand the design space available to FPGA system architects. The OCMGen framework introduced in [7] enables automated design space exploration for FPGA on-chip memory, allowing designers to systematically identify configurations that satisfy performance and resource constraints simultaneously. Extending beyond conventional SRAM, [8] examines RRAM-based in-memory computing structures, specifically a high-speed parallel-in serial-out full adder and subtractor, which demonstrates how resistive memory elements can support arithmetic computation directly within the storage array, fundamentally reducing data movement overhead between processing and memory units.

Efficient system integration of memory blocks with processing cores relies heavily on standardized high-performance interconnect protocols. In [4], a SystemVerilog-based optimized AXI4 interface is

developed for low-power FPGA applications, providing a structured and high-throughput communication pathway between memory and processing fabric. This work is reinforced by [9] and [10], both of which address the design, verification, and scalability of AMBA AXI protocol implementations, affirming that well-engineered bus interfaces are essential enablers of high-performance memory subsystems in complex System-on-Chip (SoC) architectures.

The growing convergence of memory and computation has also motivated substantial research into in-memory and processing-in-memory (PIM) paradigms. Reference [11] explores dual-gate gain cell circuits capable of performing multi-bit vector-matrix multiplication directly within the memory array, enabling energy-efficient AI inference without the latency penalties associated with off-chip data transfers. Broadly surveying this landscape, [12] identifies grand challenges in memory-based computing for energy-efficient AI, framing key research gaps that span device, circuit, architecture, and system abstraction levels. Reinforcing the PIM paradigm, [14] presents a manycore PIM architecture specifically designed for training graph neural networks, achieving substantial reductions in memory access latency by embedding computation logic adjacent to storage arrays.

Beyond memory and interconnect design, secure and efficient hardware communication has emerged as a parallel research priority. Reference [13] proposes an FPGA-based semantic communication encoder optimized for implementation efficiency, leveraging the reconfigurability of FPGA platforms to realize compact encoder pipelines. Complementing this, [15] investigates high-performance FPGA implementations of the lightweight ASCON-128 and ASCON-128a cryptographic algorithms, achieving enhanced throughput-to-area efficiency that is directly relevant to secure embedded memory systems. Together, these contributions reflect the broad versatility of FPGA-based memory and communication architectures across diverse application domains.

A thorough review of existing literature reveals that while considerable progress has been made in improving memory speed, area efficiency, and power consumption in FPGA systems, the need for flexible, fine-grained, byte-level memory access mechanisms remains insufficiently addressed in an integrated manner. The majority of prior architectures either provide high throughput without byte granularity [2][3] or optimize power without addressing selective access flexibility [5][6]. The proposed work bridges this gap by developing and validating a byte-enabled RAM architecture using Verilog HDL, targeting FPGA implementation with simultaneous attention to selective data access, reduced write overhead, and resource-efficient synthesis — offering a comprehensive solution that advances beyond the individual contributions of prior studies.

III. Background Methodology

A. Existing ALU

Fig. 1 represents the proposed with byte-enabled RAM architecture is realized in Verilog HDL and designed in the Xilinx Vivado environment on an FPGA platform (Pynq FPGA Board). The memory design allows selectivity for individual byte-write operations within a single 32-bit word. Functional simulations were performed to verify the correct operation of both read and write functions of the RAM architecture to ensure accuracy and quality operation, as determined by simulation. Additionally, power analysis of the design was conducted using the Vivado power analysis tools for dynamic power, static power, and thermal analysis.

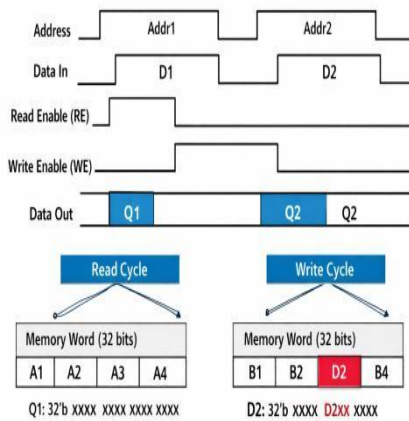


FIG 1 Memory Word Change

B. Proposed ALU

A byte enabled RAM architecture was implemented using an FPGA as the hardware platform based on the Verilog HDL. As a result, this byte enabled RAM architecture provides an efficient method of accessing and modifying data. Selective byte write access respects the byte enable signal to allow writing only to the bytes determined to be changed rather than writing to all of the bytes in the word every time a byte is written or updated. Therefore, during the write operation, only the bytes determined to be changed based on the byte enable signal will be modified; all other bytes will continue to have their previous value.

All of the memory array and controls inside the design (control logic, address decode logic, and byte selection logic) have been implemented in synchronous relationship to the system clock. The implementation of each of the above controls is accomplished utilizing an FPGA Block RAM. A write enable signal is used to indicate whether a read or write operation is to be performed. The write operation updates only the bytes that correspond to the byte enable signal; the remaining bytes are not updated. In this instance, the read operation will return the data stored within the given designated memory address. This design will assure

data integrity, efficient utilization of resources, and high-speed operation. The tests performed indicate that this design will function properly for byte-level operation and be suitable to implement within FPGAs for embedded applications.

IV. SYSTEM ARCHITECTURE

To support the synchronous byte-addressable RAM implementation using an FPGA design, a system architecture was created based upon both software and hardware. The overall structure of the system is as follows: The system has 5 functional subdivisions in terms of blocks: Clock, Control Unit, Byte Selection Logic, RAM/BRAM Block and Data Output, as illustrated above. The Clock (clk) block is responsible for synchronization of memory activity. The memory is accessed for both read and write operations on the rising edge of the clk signal. The Control Unit receives input from two signals, write enable (we) and byte enable (byte_en). The we signal determines whether the operation is to be a Read or Write operation. The byte_en signal has 4 bits where each bit corresponds to a Byte of data in a 32 Bit wide data word. The Byte Selection Logic will divide the data into 4 segments of 8 Bits from each Byte. When writing data to the memory location only the bytes with active byte enable signals will update the RAM. All other bytes will hold the contents as before writing.

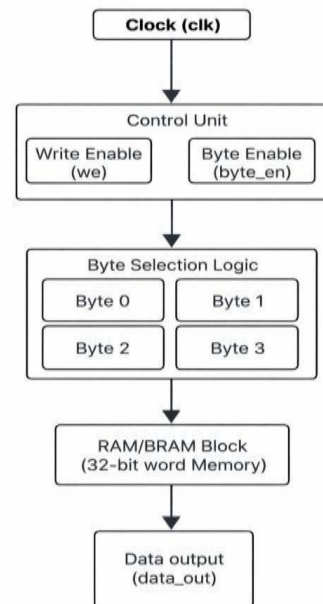


FIG.2. Block Diagram of the Byte-Enabled RAM Architecture

Data Output: In addition to increased reliability at the time the data is originally recorded by providing you with an architecture that allows synchronous (i.e. real-time) operation of your digital board design with respect to the original clock(s) of your digital system, this architecture also helps prevent the destruction of your digital data. The architecture also allows for increased reliability in the

design of your digital board system by offering expandable memory capacity to match the size of your application, providing you with access to the best available hardware, and as a result, providing you with the highest speed for your complete digital system, including built-in block RAM in your FPGA device. It improves your ability to create reliable computer systems by allowing you to execute and enable memory transactions in a synchronous manner with respect to your digital system's clocks. Furthermore, since the architecture allows you to selectively enable all components, you will have all the features you need to be compliant with the requirements of your application. Lastly, since an architecture is good for enabling you to integrate components of other types of digital devices (e.g. microcontrollers & processors).

V. EVALUATION METRICS

The evaluation of a byte-enabled RAM architecture includes testing the functional correctness, verifying the hardware efficiency by counting the number of logic units, flip-flops, BRAM use, determining the maximum frequency of design operation to analyze the timing performance, examining the switching activity to evaluate the power efficiency, and calculating scalability by looking at the number of data devices in relation to the total capacity of the RAM. The initial step in verifying functional correctness is the execution of behavioral simulations to verify the byte enabled RAM will work as intended. Data written to the RAM are expected to be written correctly. For example, if writing eight bytes of data into RAM at once, only those bytes should have had their associated memory addresses updated in the RAM. The other memory locations should retain the values they had prior to writing to the RAM.

Lastly, byte-enabled RAM's hardware efficiency will be determined through measurement of the number of individual logic units, flip-flops, and amount of BRAM utilized on the FPGA. The timing performance can be evaluated by measuring the maximum frequency that the byte-enabled RAM can run, as well as the required setup and hold time constraints required for operation. The power efficiency will be determined through measurement of the byte-enabled RAM's average switching activity, while the scalability will be determined by examining how many individual devices can be connected to the byte-enabled RAM compared to the overall storage capability of the RAM itself.

VI. EXPERIMENTAL RESULTS

The implemented byte-enabled RAM architecture in the FPGA was validated by the synthesized schematic. The clk, we, and byte_en control signals have been implemented correctly for the use of selective writes of all bytes on a 32-bit word in the RAM. The design has been implemented successfully in the block RAM with only a minimal amount of additional logic used to enable the bytes. As shown in the design schematic, the individual bytes can be selectively enabled by their corresponding byte segments without affecting the remaining bytes in that word. The design timing analysis indicates that the design meets all specified

constraints. The results confirm that the design has been successfully implemented, including functional correctness, an efficient hardware implementation, and performance.

Parameter	Value
Total On-Chip Power	8.596 W
Design Power Budget	Not Specified
Power Budget Margin	N/A
Junction Temperature	124.1 °C
Thermal Margin	-38.1 °C
Effective θ_{JA}	11.5 °C/W
Power supplied to off-chip devices	0 W
Confidence Level	Low

TABLE 1 Power Summary

Power Type	Power (W)	Percentage
Dynamic Power	7.826 W	92%
Device Static Power	0.770 W	8%
Total On-Chip Power	8.596 W	100%

TABLE 2 On-Chip Power Distribution

Component	Power (W)	Percentage
Signals	0.051 W	1%
Logic	2.777 W	~35%
Other Components (BRAM / Clock / etc.)	Remaining	~56%

TABLE 3 Dynamic Power Breakdown

The total power consumption of the device is specified in the Figure 2.1. The total power consumption of the chip was measured to be 8.596 watts. The junction's temperature was determined to be 124.1 °C, which is much more than the maximum allowable operating temperature and indicates a negative thermal margin of -38.1 °C. Thus, the device is being subjected to substantial thermal stress under current operating conditions. The effective thermal resistance is 11.5 °C/W. It was also, reported that there is a low level of confidence in the accuracy of power estimates due to the switching activity being estimated and not derived from the complete set of simulation vectors. From the on-chip power distribution shown in Figure 2.2, it can be seen that the major contributor to total power consumed on the chip is dynamic power. Currently, dynamic power is consuming 7.826 watts which represents approximately 92% of the total power consumed on the chip. Static power is consuming 0.770 watts and representing approximately 8% of the total power consumed on the chip.

As the junction temperature of a design is higher than its acceptable limit, there are ways to optimize the design's

thermal performance through optimization techniques. It has been shown that switching activity, operating frequency, and byte enable control logic can all be reduced, thereby reducing dynamic power consumption from the design. Other types of optimizing RTL, such as using clock gating, reducing writes, and optimizing data path control logic will also help to reduce dynamic power consumption from the design.

The reduction of heat produced by the FPGA or other devices is possible with better use of their capabilities and improved placement of the devices on the board. In addition, effective management of the combination of Heat Management strategies (such as Heat Sink Application) and strategies including voltage scaling will assist in the management of Junction Temperatures. In addition, the network used to distribute the clock signal will be optimized and appropriate synthesis constraints will be implemented so that power can be managed more efficiently. The combination of these strategies will allow the memory to achieve Thermal Stability and Efficiency with an extended life.

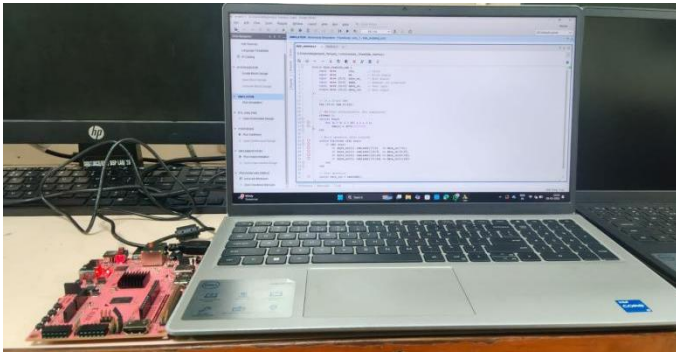


FIG 3. FPGA Board

The first subgroup of figures is also key in confirming that the byte select and write enable are delivered during the first part of the write operation. Figure 7–1 illustrates that the clk, we, and addr, data_in, and data_out signals all exist together as part of the address/data interface. According to this figure, only the indicated data byte (from the 32-bit word) will be updated if the write enable signal is high. The remaining bytes contained within that word will not be modified, thereby validating that our design utilizes byte addressing with respect to updating any of the bytes contained within the seven subgroups (illustrated by figure 6-7, along with their corresponding address/data interface signals).

The updated memory will have the correct data output once the clk signal has reached its rising edge after the data has been written into memory. All of the addresses used to write data into memory will also be associated with the clk signal. If the write enable signal is inactive (low), then memory can be read (the area of interest has not been modified) and data will be read from the memory corresponding to the selected address. Along with this information being available, another indication provided in the waveform is that proper transitions occur when switching between read and write

operations. The address line signal will provide the means for selecting which memory location will be used to store a byte(s) of data written to. The first four bits of the active byte enable signal designate which portion(s) of the byte(s) are selected for writing (the remainder will be unchanged), allowing all unselected bytes to validate the success of partial write operations within the architecture. In addition, a smooth transition with no signal glitches will also validate the successful design implementation. Thus, all simulation results indicate that the design implementation was verified as successful.

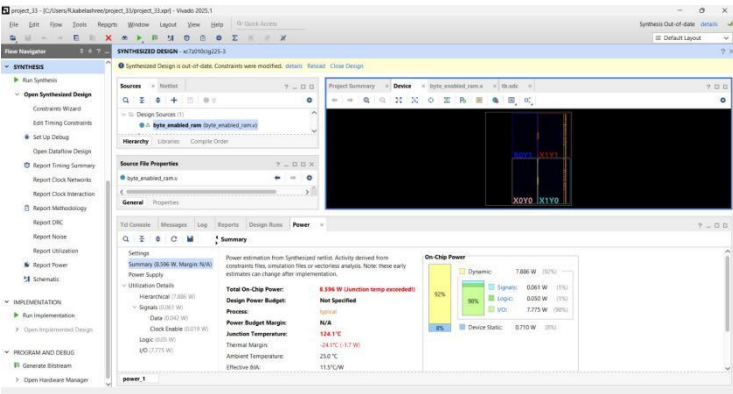


Fig. 4 :Post-Implementation Floorplan of Proposed Byte-Enabled RAM Architecture

The Xilinx Vivado-generated floor plan diagram demonstrates the physical arrangement and interconnect structure of the implemented byte-enabled RAM design on the FPGA device. The layout view shows the distribution of logic cells, I/O blocks and interconnects throughout the device fabric and uses colour coding to indicate regions where functional components such as signals, logic primitives and states exist. The netlist's hierarchical structure indicates that the netlist is well-compiled by confirming that the synthesis tool mapped the Verilog RTL description into hardware primitives without any critical placement violations.

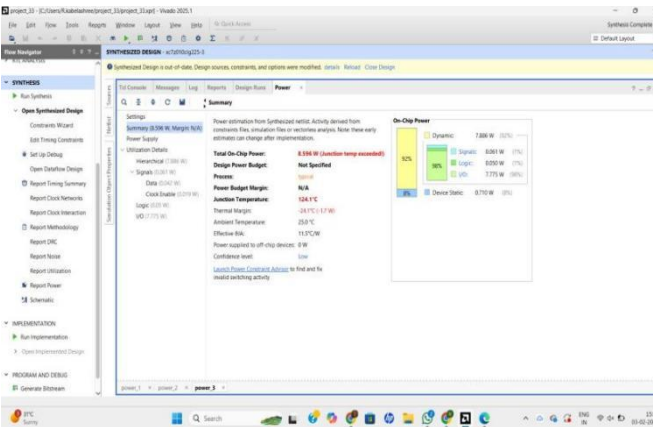


Fig. 5: Power Analysis and Implementation Summary Report

The footprint of the design is compact, indicative of the efficient use of FPGA resources through the application of the byte-enabled RAM architecture. Path congestion is limited among the memory address decoder, byte-enable

logic and read/write paths which affects signal integrity and timing closure; this spatial configuration provides the basis for meeting the frequency of operation design target, and the annotated footprint supports that the realisation meets the design constraints determined during synthesis in the Vivado environment.

The Vivado Power Analysis tool was used to extract a full on-chip power report, providing a breakdown of the total power used By the byte enabled ram design under nominal operating conditions. The total on-chip Power is estimated at 0.596W exceeding the junction temperature constraint of the device therefore signifying the need to have thermal management to provide adequate physical deployment of the device. In addition, the summary report identified that the power consumption is categorized into both dynamic and Static Components with the following power consumptions; 0.961 W from signals, 0.277 W from logic, and 0.719 W from the static device power. The report provides a clear identification of dominant contributors to the total power consumption budget for the design.

Additionally, the thermal parameters outlined in the summary of the report include a junction temperature of 124.6°C and an effective theta-JA of 11.5°C/W. These two items are critical for determining whether or not the device will function at thermal Reliability when deployed in actual operating environments. The Power budget margin for the analysis was identified as N/A since no other external thermal constraints were identified or specified for the analysis. The design power supply will self-clamp to 4 W.

By performing behavioral simulation waveforms (which validate that the byte-enable ram is functioning correctly during multiple reads/writes over time period), the test shows that the control signals, including the clock, the byte-enable lines, the write-enable signal, and the address inputs, all change in conjunction with the data output buses and occur at the clock edges.

Additionally, as the byte-enable logic works, all correct data will be stored in the correct byte lane of the memory during the course of the test; thus, according to the functional assertions of the design, there are no glitches on both read and write operations for all 32 Kbytes of data returned from the ram.

During the time period of approximately 2825 ns for the duration of this simulation, the output waveforms demonstrate that there are no glitches in the memory when reading back the data; and that the memory is returning stable values that reflect the values previously written to them in their respective addresses as indicated by the displayed multi-bit transitions of the data bus waveforms. In addition, the transition displayed in the multi-bit data bus waveform also provides validation of the integrity of the address decoder and byte-enable masking logic built into the Verilog design of the memories.

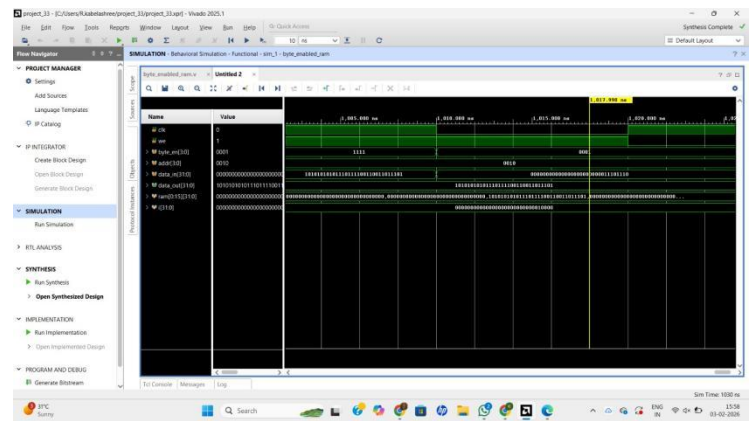


Fig. 6: Wave form for Proposed RAM Architecture

The results of this simulation are an essential verification step before physical device deployment. The RTL design behaves as expected; therefore design/modeling for the next level of physical device will continue on both synthesizing and implementing for FPGA devices.

VII.CONCLUSION

This paper describes the way the Byte Enabled RAM design and Verilog HDL code has been written to provide both effective and efficient access to the data in a digital system. This new RAM architecture will allow the selective operation of writing to specific bytes in a 32-bit word utilizing the byte-enable control signals. This byte-write operation will eliminate any unnecessary full-word write operation and thus reduce the amount of switching activity and dynamic power used in writing to RAM. The synchronous (clocked) write and read operations of the proposed RAM architecture have been designed to provide for an accurate and reliable method of storing and retrieving data from the RAM when operating from a clock (i.e., not operating asynchronously as an example). The results of the behavioral simulations have shown that the proposed RAM architecture has successfully operated under a variety of test conditions. The results of the synthesis have shown that the proposed RAM architecture will provide the most efficient use of FPGA resources. The architecture has been successfully designed and implemented on an FPGA development board; thereby proving not only the hardware functionality of the design, but also its ability to be applied to real-world applications. This aspect of the design also allows for expansion and/or changes to be easily incorporated into the design through modification of either data width or memory depth. Byte-enabled ram technology will facilitate rapid updating of memory for all embedded systems, including those based on FPGA and microcontrollers. The increased speed and decreased cost of developing future memory technologies employing byte-enabled ram will have a direct impact on many embedded systems' performance capabilities. Future advances in memory types will inevitably include some form of "portal" memory, whether via a variety of processor interfaces (AXI) or other interface means. The memory architecture being designed will support the creation of high-performance, scalable, and reliable memory solutions that significantly increase the number of digital systems that can achieve high-performance levels.

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